Practical No 5

5. Write a program for the Tic-Tac-Toe game

```
import random


# Initialize empty board
board = [' '] * 9


# Function to display the board
def display():
    print(f"{board[0]} | {board[1]} | {board[2]}")
    print("--+---+--")
    print(f"{board[3]} | {board[4]} | {board[5]}")
    print("--+---+--")
    print(f"{board[6]} | {board[7]} | {board[8]}")
    print()


# Function to check win condition
def check_win(player):
    winning_combinations = [
        (0,1,2),(3,4,5),(6,7,8),   # rows
        (0,3,6),(1,4,7),(2,5,8),   # columns
        (0,4,8),(2,4,6)            # diagonals
    ]
    return any(board[a] == board[b] == board[c] == player for a,b,c in winning_combinations)


# Function for user move
```

```
def user_move():
    while True:
        try:
            move = int(input("Enter position (1-9): ")) - 1
            if move < 0 or move > 8:
                print("Invalid position! Choose 1-9.")
            elif board[move] != ' ':
                print("Position already filled!")
            else:
                board[move] = 'X'
                break
        except ValueError:
            print("Invalid input! Enter a number between 1 and 9.")

# Function for computer move
def computer_move():
    empty = [i for i in range(9) if board[i] == ' ']
    if empty:
        move = random.choice(empty)
        board[move] = 'O'

# Main game loop
def play_game():
    print("Welcome to Tic-Tac-Toe! You are 'X', Computer is 'O'.\n")
    display()
    while True:
        # User move
```

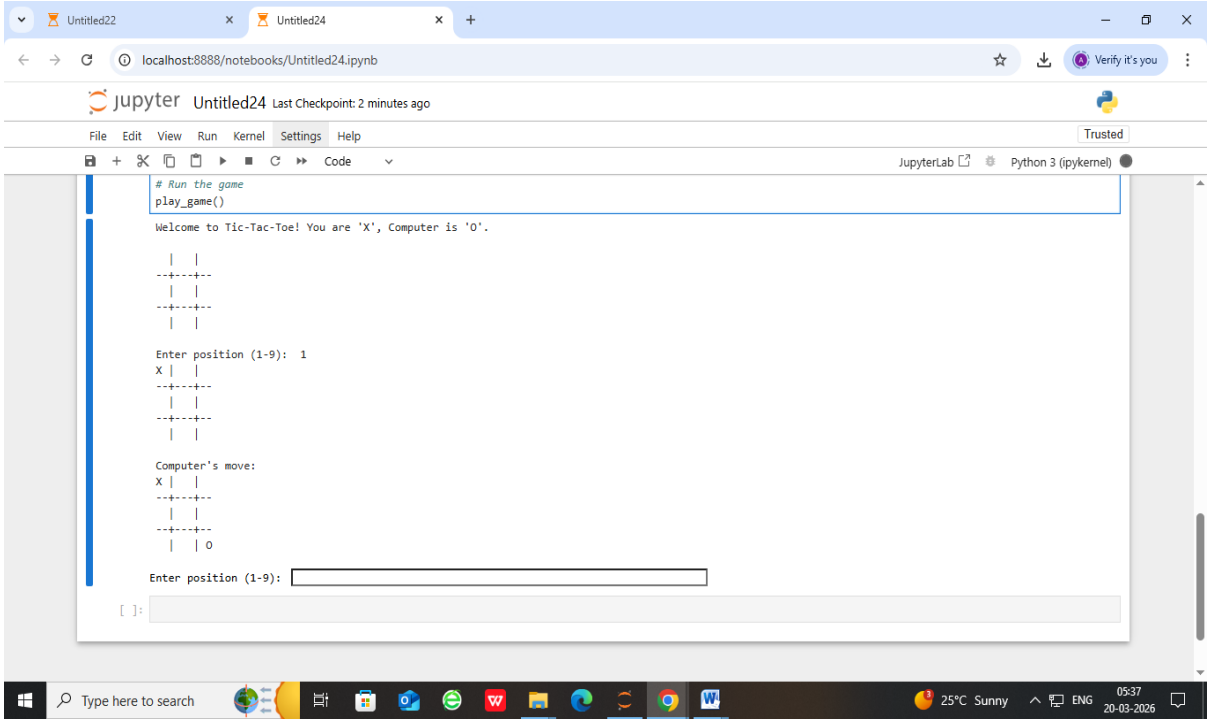
```
        user_move()
        display()
        if check_win('X'):
            print("You Win! 🎉")
            break

        if ' ' not in board:
            print("Draw! 🤝")
            break

        # Computer move
        print("Computer's move:")
        computer_move()
        display()
        if check_win('O'):
            print("Computer Wins! 🤖")
            break

        if ' ' not in board:
            print("Draw! 🤝")
            break

# Run the game
play_game()
```


Untitled22
Untitled24
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Code
JupyterLab
Python 3 (ipykernel)
# Run the game
play_game()
Welcome to Tic-Tac-Toe! You are 'X', Computer is 'O'.
| |
--+---+--
| |
--+---+--
| |
Enter position (1-9): 1
X | |
--+---+--
| |
--+---+--
| |
Computer's move:
X | |
--+---+--
| |
--+---+--
| | O
Enter position (1-9):
[ ]:
Type here to search
25°C Sunny
ENG
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